

**FIITJEE**  
**ALL INDIA TEST SERIES**  
**JEE (Advanced)-2022**  
**FULL TEST – XI**  
**PAPER –1**  
**TEST DATE: 23-08-2022**

**Time Allotted: 3 Hours**

**Maximum Marks: 180**

**General Instructions:**

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.  
**Section – A (01 – 04, 20 – 23, 39 – 42):** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.  
**Section – A (05 – 10, 24 – 29, 43 – 48):** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).  
**Section – B (11 – 13, 30 – 32, 49 – 51):** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.  
**Section – C (14 – 19, 33 – 38, 52 – 57):** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**MARKING SCHEME**

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                     |
|----------------|---|----|---------------------------------------------------------------------|
| Full Marks     | : | +3 | If ONLY the correct option is chosen.                               |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | -1 | In all other cases.                                                 |

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

|                |   |    |                                                                                                     |
|----------------|---|----|-----------------------------------------------------------------------------------------------------|
| Full Marks     | : | +4 | If only (all) the correct option(s) is (are) chosen;                                                |
| Partial Marks  | : | +3 | If all the four options are correct but ONLY three options are chosen;                              |
| Partial marks  | : | +2 | if three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks  | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option;        |
| Zero Marks     | : | 0  | If none of the options is chosen (i.e. the question is unanswered);                                 |
| Negative Marks | : | -2 | In all other cases.                                                                                 |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

|            |   |    |                                         |
|------------|---|----|-----------------------------------------|
| Full Marks | : | +4 | If ONLY the correct integer is entered; |
| Zero Marks | : | 0  | In all other cases.                     |

**Section – C:** Answer to each question will be evaluated according to the following marking scheme:

|            |   |    |                                                                         |
|------------|---|----|-------------------------------------------------------------------------|
| Full Marks | : | +2 | If ONLY the correct numerical value is entered at the designated place; |
| Zero Marks | : | 0  | In all other cases.                                                     |

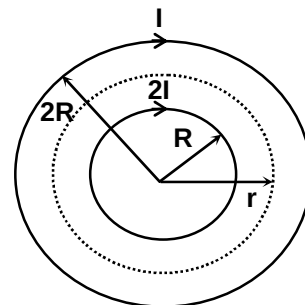
# Physics

## PART – I

### Section – A (Maximum Marks: 12)

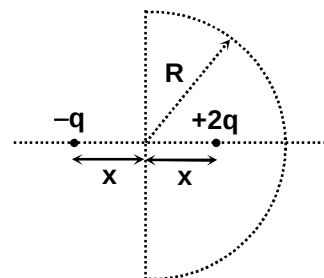
This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

1. A long solenoid contains another coaxial solenoid. Their radii are  $2R$  and  $R$  respectively. The two solenoids have the same number of turns per unit length and initially both carry no currents. At the same instant current starts increasing linearly with time in both solenoids. At any moment the current flowing in the inner solenoid is twice as large as that in the outer one and their directions are same. As a result of the increasing currents a charged particle, initially at rest between the solenoids, starts moving along a circular trajectory as shown in the figure. What is the radius  $r$  of this circle?



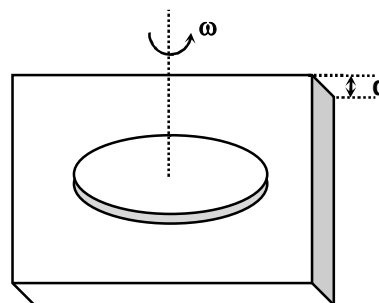
- (A)  $\sqrt{\frac{3R}{2}}$   
 (B)  $\sqrt{2}R$   
 (C)  $\sqrt{3}R$   
 (D)  $\frac{3R}{2}$

2. Consider a hemi-spherical closed Gaussian surface of radius  $R$ . Two point charges  $+2q$  and  $-q$  are placed at a distance  $x$  from the plane surface as shown in the figure on the line passing through the centre and perpendicular to the plane surface. If the electric flux on the plane and curved surface are equal, find the value of  $x$ .



- (A)  $\frac{R}{2}$   
 (B)  $\frac{R}{\sqrt{2}}$   
 (C)  $\frac{2R}{3}$   
 (D)  $\frac{R}{2\sqrt{2}}$

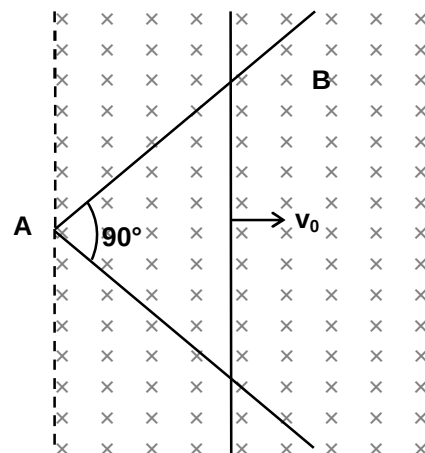
3. A disc of radius  $R$  which is brought in contact with an oil film on a horizontal surface is rotated with a constant angular velocity  $\omega$  about its vertical axis. The coefficient of viscosity of oil is  $\eta$  and the thickness of the oil film is  $d$ . Find the power delivered by the external agent to rotate the disc. (Assume velocity gradient to be uniform)



- (A)  $\frac{\pi\omega^2\eta R^4}{2d}$   
 (B)  $\frac{\pi\omega^2\eta R^4}{d}$   
 (C)  $\frac{\pi\omega^2\eta R^2}{2d}$

(D)  $\frac{2\pi\omega^2\eta R^4}{d}$

4. A conducting uniform wire bent into the shape of a right angle is held stationary on a horizontal frictionless surface. A very long uniform thin conducting rod of mass  $m$  projected with velocity  $v_0$  from the apex A of the bend wire. The resistance per unit length of the wire and the rod is  $\lambda$ . The entire arrangement is placed in a uniform magnetic field  $B$  perpendicular to the plane formed by the wire and the rod. Find the distance travelled by the rod before it becomes stationary.

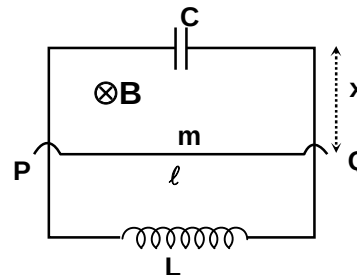


- (A)  $\sqrt{\frac{m\lambda v_0(\sqrt{2}+1)}{B}}$   
 (B)  $\frac{\sqrt{m\lambda v_0(\sqrt{2}+1)}}{2B}$   
 (C)  $\frac{\sqrt{m\lambda v_0(\sqrt{2}+1)}}{B}$   
 (D)  $\sqrt{\frac{m\lambda v_0(\sqrt{2}+1)}{2B}}$

#### Section – A (Maximum Marks: 24)

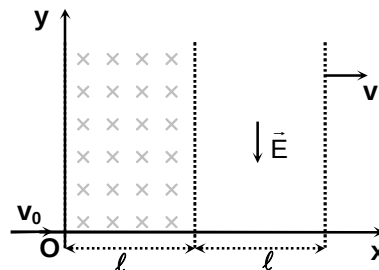
This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

5. A conducting wire PQ of mass  $m$  and length  $\ell$  can slide freely on a pair of frictionless vertical conducting rails. A uniform magnetic field  $B$  exists in the entire region directed normally inwards to the plane of the rails. The rails are connected at the top end by a capacitor of capacitance  $C$  and at the bottom by an inductor of inductance  $L$  as shown in the figure. The conductor is released from the top (i.e.,  $x \approx 0$ ) from rest at time  $t = 0$ . The acceleration due to gravity is  $g$ . Neglect the resistance of the rails and the conductor. Then



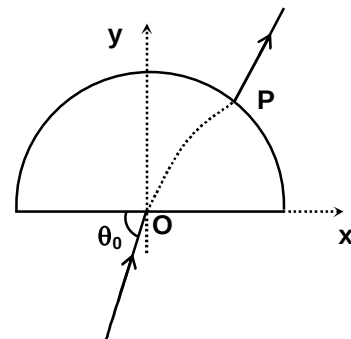
- (A) the conducting wire will be in equilibrium at  $x = \frac{mgL}{B^2\ell^2}$   
 (B) the current in the conducting wire as a function of  $x$  is  $I = B\ell Ca + \frac{2B\ell}{L}x$ , where  $a$  is the instantaneous acceleration of the wire PQ.  
 (C) the conducting wire will be in equilibrium at  $x = \frac{mgL}{2B^2\ell^2}$   
 (D) the current in the conducting wire as a function of  $x$  is  $I = B\ell Ca + \frac{B\ell}{L}x$ , where  $a$  is the instantaneous acceleration of the wire PQ.

6. There is a uniform magnetic field  $\vec{B} = -B_0 \hat{k}$  from  $x = 0$  to  $x = \ell$ . In the region from  $x = \ell$  to  $x = 2\ell$  there is a uniform electric field  $\vec{E} = -E_0 \hat{j}$ . A positively charged particle of charge  $q$  and mass  $m$  moving along  $x$ -axis enters the region of magnetic field with velocity  $\vec{v} = v_0 \hat{i}$ . After passing through the region of magnetic and electric field it emerges parallel to its initial direction of motion with speed  $v$  as shown in the figure. Then (given  $\ell = \frac{mv_0}{2qB_0}$ )



- (A)  $\frac{E_0}{B_0} = \frac{3v_0}{4}$   
 (B)  $\frac{E_0}{B_0} = \frac{\sqrt{3}v_0}{2}$   
 (C) Total time of its motion in the regions of electric and magnetic fields is  $(\pi + 2\sqrt{3}) \frac{m}{6qB_0}$   
 (D)  $v = \frac{v_0}{2}$

7. A hemisphere made of a transparent material of variable refractive index has radius 12 cm and centre at origin O. The refractive index of the material of the sphere varies as  $\mu = \frac{24}{24-x}$ , where  $x$  is in cm. A ray of light is incident at the point O at an angle  $\theta_0$  with the  $x$ -axis and comes out through a point P on its curved surface. Given  $\theta_0 \approx 90^\circ$ , then



- (A) path of the ray inside the sphere is parabolic  
 (B) path of the ray inside the sphere is circular  
 (C)  $x$ -coordinate of point P is 3 cm  
 (D)  $x$ -coordinate of point P is 1.50 cm

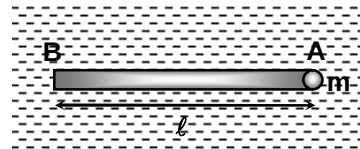
8. Two stars of masses  $m$  and  $2m$  are at a distance  $\ell$  apart. They are moving in circular paths about the centre of mass of the system due to mutual gravitational force. Then

- (A) ratio of their kinetic energy w.r.t. the centre of mass frame is 2:1.  
 (B) ratio of their angular momenta w.r.t. centre of mass frame is 4 : 1

- (C) the total angular momentum of the system w.r.t. centre of mass frame is  $\sqrt{\frac{4Gm^3\ell}{3}}$

- (D) total kinetic energy of the system in centre of mass frame is  $\frac{Gm^2}{\ell}$

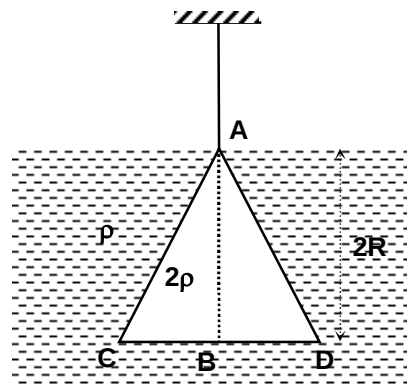
9. A point mass  $m$  is rigidly fixed to one end A of a uniform thin rod of mass  $m$  and density  $\rho$ . The rod AB is released from rest inside a liquid of density  $2\rho$  from the horizontal position. Then, just after the release



- (A) acceleration of point A is  $\frac{3g}{5}$  downward  
 (B) acceleration of point B is  $\frac{9g}{5}$  upward

- (C) acceleration of mid point of rod is  $\frac{3g}{4}$  upward
- (D) time taken by the rod to become vertical (i.e. rotate through an angle  $90^\circ$ ) is  $\sqrt{\frac{5\pi\ell}{12g}}$

10. A right circular solid cone of radius  $R$ , height  $2R$  and density  $2\rho$  suspended by a string is just immersed completely in a fluid of density  $\rho$ . Neglect atmospheric pressure then

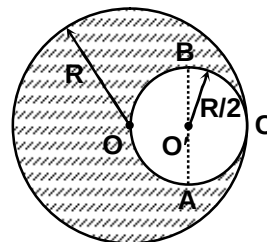


- (A) Net vertical force due to hydrostatic pressure on the curved surface of the half part of the cone is  $\frac{8}{3}\rho g R^3$
- (B) Net horizontal force due to hydrostatic pressure on the half part of the cone ACB is  $\frac{8}{3}\rho g R^3$
- (C) The horizontal force exerted by the half of the cone ACB on the other half of the cone is  $\frac{2}{3}\rho g \pi R^3$
- (D) Tension in the string is  $\frac{2}{3}\rho g \pi R^3$

### Section – B (Maximum Marks: 12)

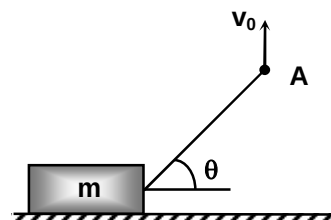
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. An infinitely long cylinder of radius  $R$  has a cylindrical cavity of radius  $\frac{R}{2}$ . The region inside the cylinder is filled with uniform charge density  $\rho$  except inside the cavity. A charged particle of charge  $q$  and mass  $m$  is projected with non-zero velocity  $v_0$  inside the cavity from point A to reach point B (AB is perpendicular to line OC). If the minimum magnitude of velocity of projection  $v_0$  of the particle required to reach point C is  $\sqrt{\frac{q\rho R^2}{n\epsilon_0 m}}$ . Find the value of  $n$ .



12. A particle of mass  $0.5 \text{ kg}$  is projected horizontally in air with initial velocity of  $9 \text{ m/s}$ . During the motion of the particle in air a constant air drag of magnitude  $5 \text{ N}$  acts on it opposite to its instantaneous direction of motion. Find its speed (in  $\text{m/s}$ ) when its direction of motion makes an angle of  $30^\circ$  with the horizontal. (given  $g = 10 \text{ m/s}^2$ )

13. A block of mass  $m = 1\text{ kg}$  is placed on a frictionless horizontal surface. One end of a massless string of length  $0.8\text{ m}$  is connected to the block. The string is straight and the other end of the string is moved vertically up with constant velocity  $v_0$ . When the string makes an angle  $\theta = 30^\circ$  with the horizontal surface the block leaves the surface. Find the speed  $v_0$  (in  $\text{m/s}$ ) (given  $g = 10\text{ m/s}^2$ )



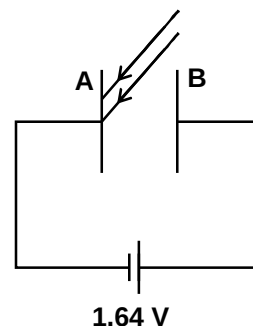
### Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

#### Question Stem for Question Nos. 14 and 15

##### Question Stem

In the arrangement shown in the figure, light of wavelength  $500\text{ nm}$  is incident on plate A of a parallel plate capacitor. The emitted photoelectrons from plate A are accelerated by a potential difference of  $1.64\text{ V}$ . These electrons strike the another plate B from which electromagnetic radiations are emitted. The minimum wavelength of emitted photons is  $620\text{ nm}$ . The work function of the metal is  $X\text{ eV}$  and the maximum kinetic energy of photoelectrons emitted from metal A is  $Y\text{ eV}$ . (Given  $hc = 1240\text{ eV-nm}$ )

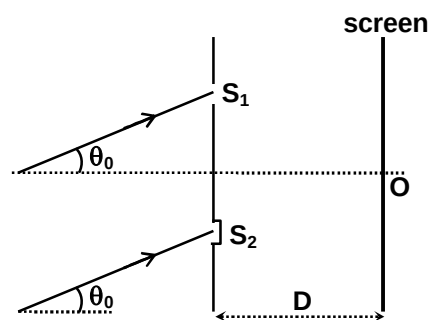


14. The value of  $X$  is.....
15. The value of  $Y$  is.....

#### Question Stem for Question Nos. 16 and 17

##### Question Stem

In Young's double slits experiment a parallel monochromatic light beam of wavelength  $500\text{ nm}$  falls on the slits at angle  $\theta_0$  with the normal to the plane of the slits. A transparent sheet of refractive index  $1.5$  and thickness  $19.75\text{ }\mu\text{m}$  is placed in front of the slit  $S_2$ . The intensity on the axis of the screen  $O$  is  $I$  and maximum fringe intensity on the screen is  $I_0$ . The value of  $\frac{I}{I_0} = X$  and the minimum distance from the origin  $O$  where the intensity is maximum is  $Y\text{ mm}$ . (Given  $\sin\theta_0 = \frac{1}{50}$ , the distance between the slits  $S_1$  and  $S_2$  =  $0.5\text{ mm}$ , the distance between the slits and the screen =  $1\text{ m}$ )



16. The value of  $X$  is.....
17. The value of  $Y$  is.....

**Question Stem for Question Nos. 18 and 19****Question Stem**

End A of a uniform rod of length  $\ell = 5\text{m}$  and cross section area  $S = 100\text{ cm}^2$  is maintained at some constant temperature. The thermal conductivity of the rod varies with distance  $x$  from end A as  $k = k_0(1 + 0.2x)$ , where  $k_0 = 8.4\text{ Wm}^{-1}\text{K}^{-1}$ . The other end of the rod is radiating energy at the rate of  $33.6\text{ J/s}$ . Assume that except the ends A and B the rod is thermally insulated. Given Wien's constant  $b = 2.9 \times 10^{-3}\text{ m-K}$ . The wavelength of radiation emitted with maximum energy density from end B is  $\lambda_0 = 7250\text{ \AA}$ . If the temperature of end A is  $X$  kelvin and temperature at the mid point of the rod is  $Y$  kelvin. Consider the end B of the rod as a black body radiator. (Take  $\ln 2 = 0.7$  and  $\ln 3 = 1.1$ )

18 The value of  $X$  is.....

19 The value of  $Y$  is.....

# Chemistry

## PART – II

### Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

20. The maximum limiting labelling of oleum is approximately?

- (A) 200%
- (B) 163.5%
- (C) 122.5%
- (D) 180%

21.  $A \xrightarrow[\text{(cyclic polymerisation)}]{\text{Red hot iron tube, 873 K}} B \xrightarrow{\text{nitration mixture}} C \xrightarrow[\text{6[H]}]{\text{Sn+HCl}} D \xrightarrow[\text{catalyst}]{\text{3Br}_2} E \xrightarrow[\text{Low temperature}]{\text{NaNO}_2 + \text{HCl}} F \xrightarrow[\text{H}_2\text{O}]{\text{H}_3\text{PO}_2} G$

$$\left( \begin{array}{l} \text{C\%} = 22.848 \\ \text{H\%} = 0.952 \\ \text{Br\%} = 76.20 \end{array} \right)$$

Molar mass of G is

- (A) 240
- (B) 300
- (C) 315
- (D) 205

22. Which one of the following statement is incorrect?

- (A) In  $\text{PF}_5$  the fluorine atoms are indistinguishable by means of spectroscopy
- (B) The exchange of fluorine atoms in  $\text{PF}_5$  is too rapid
- (C) The two TBP arrangements are related to each other by simple rotation, the entire process is called inversion reaction
- (D) In  $\text{PF}_5$  ground state TBP is converted into SP transition state and back to new TBP structure

23. If one mole of a ideal gas  $\left( C_p = \frac{5R}{2} \right)$  is expanded isothermally at 300 K until its volume is tripled and  $P_{\text{ext}} = 0$  then which one of the following option is correct?

- (A)  $W \neq 0$   $\Delta S = \text{infinity}$
- (B)  $W = 0$   $\Delta S = R \ln 3$
- (C)  $\Delta E = 0$   $\Delta S = 0$
- (D)  $\Delta E = 0$   $\Delta S = \text{infinity}$

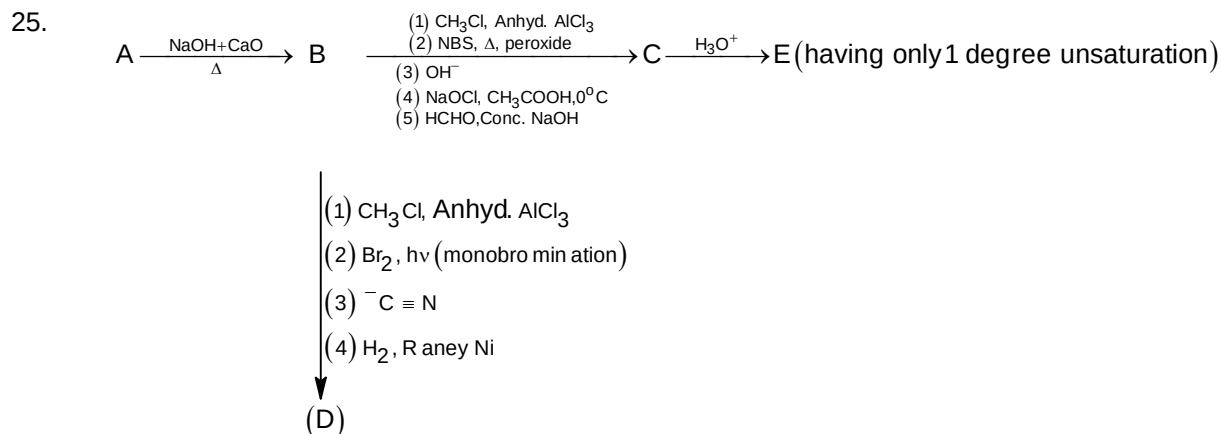
### Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

24. Which of the following option is/are correct?

- (A)  $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$  (Density)
- (B)  $\text{Li} < \text{Na} < \text{K} < \text{Rb} < \text{Cs}$  (Re activity toward water)
- (C)  $\text{Li} < \text{Na} < \text{I}^- < \text{Ag}$  (Reducing strength in aqueous phase)
- (D)  $\text{Cs}^+ < \text{Rb}^+ < \text{K}^+ < \text{Na}^+ < \text{Li}^+$  (Hydration enthalpy)





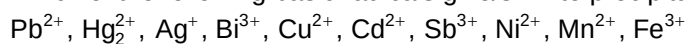
Select the correct option for above information

- (A) E gives positive Tollen's test  
 (B) D gives positive carbyl amine test  
 (C) A and C are same compound  
 (D) C is one of the product of self Cannizzaro of HCHO
26. Select the correct option  
 (A)  $\text{VO}_4^{3-}$ ,  $\text{CrO}_4^{2-}$  and  $\text{MnO}_4^-$  are isoelectronic ions  
 (B)  $\text{VO}_4^{3-}$ ,  $\text{CrO}_4^{2-}$  and  $\text{MnO}_4^-$  have intense charge transfer transition  
 (C)  $\text{MnO}_4^-$  is intensely purple due to LMCT  
 (D)  $\text{VO}_4^{3-} < \text{CrO}_4^{2-} < \text{MnO}_4^-$  (wavelength of charge transfer transition)
27. Select the correct match for refining process  
 (A) Electrolytic refining for Zinc  
 (B) Vapour phase refining for Nickel  
 (C) Zone refining for Gallium  
 (D) Vapour phase refining for Titanium
28. Select the correct radiometric age determination method/s which can be used to date minerals  
 (A)  $^{238}\text{U} - ^{206}\text{Pb}$  method  
 (B)  $^{235}\text{U} - ^{207}\text{Pb}$  method  
 (C)  $^{40}\text{K} - ^{40}\text{Ar}$  method  
 (D)  $^{87}\text{Rb} - ^{87}\text{Sr}$  method
29. A glass tube AD of uniform cross section of length 100 cm contains two columns of ideal gas AB and CD, respectively, a column of mercury of length 20 cm. When the tube is horizontal, AB = 20 cm and CD = 60 cm. When the tube held vertically with the end A up, the mercury column moves down 10 cm.  
 Which of the following statement is correct for the tube?  
 (A) the pressure of both the columns, when the tube was in horizontal position, is 37.5 cm Hg  
 (B) only D end is open  
 (C) both the ends are open  
 (D) both the ends are closed

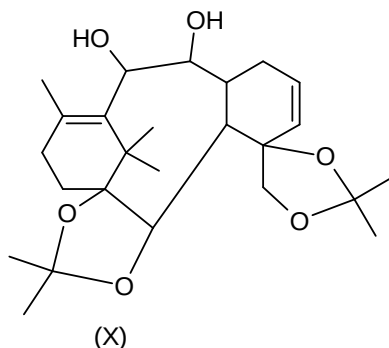
**Section – B (Maximum Marks: 12)**

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. Which of the following basic radicals give/s white precipitate with NaOH?



31.



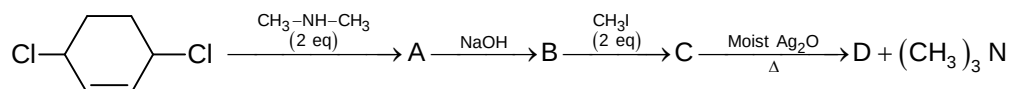
Degree of unsaturation in 'X' is equal to a; and if  $b = \frac{a \times 10}{2}$ . Then what is the value of b?

32. Which of the following are negatively charged sol?

Methylene blue,  $\text{TiO}_2$ ,  $\text{As}_2\text{S}_3$ ,  $\text{CdS}$ , Eosin, Congo red, Gelatin, Starch,  $\text{Sb}_2\text{S}_3$

**Section – C (Maximum Marks: 12)**

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**Question Stem for Question Nos. 33 and 34**
**Question Stem**


For the above reaction sequence  $x = \frac{\text{Molecular wt. of D}}{0.5}$  and

$$y = \frac{[2n + 2] - [\text{number of hydrogens in D}]}{2}$$

Here n = number of carbon atoms in D.

33. The value of x is \_\_\_\_\_?

34. What is the value of y \_\_\_\_\_?

**Question Stem for Question Nos. 35 and 36**

**Question Stem**

$\text{Cl}_2\text{C} = \text{CCl}_2, \text{SO}_2, \text{DDT}, \text{Pb}, \text{O}_3, \text{PAN}, \text{NO}, \text{CF}_2\text{Cl}_2, \text{CO}_2, \text{CH}_4$

Here,  $x = \text{number of soil pollutants} \times \text{number of water pollutants}$

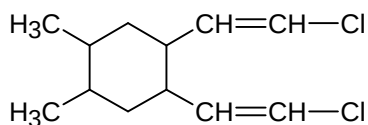
$$y = \frac{\text{number of stratospheric pollutants}}{2}$$

35. What is the value of  $x$  \_\_\_\_\_?

36. What is the value of  $y$  \_\_\_\_\_?

**Question Stem for Question Nos. 37 and 38**

**Question Stem**



For the above compound  $x = \frac{\text{number of stereoisomers}}{100}$

$$y = \frac{\text{number of optical isomers}}{0.01}$$

37. What is the value  $x$  \_\_\_\_\_?

38. What is the value of  $y$  \_\_\_\_\_?

# Mathematics

## PART – III

### Section – A (Maximum Marks: 12)

This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

39. What is the remainder when  $2^{1990}$  is divided by 1990  
 (A) 1018  
 (B) 1020  
 (C) 1022  
 (D) 1024
40. I have 6 friends and during a vacation I met them during several dinners. I found that I dined with all the 6 exactly on 1 day; with every 5 of them on 2 days; with every 4 of them on 3 days; with every 3 of them on 4 days; with every 2 of them on 5 days. Further every friend was present at 7 dinners and every friend was absent at 7 dinners. How many dinners did I have alone  
 (A) 1  
 (B) 2  
 (C) 3  
 (D) 4
41. The value of  $4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right)$  is  
 (A)  $\frac{\pi}{2}$   
 (B)  $\frac{\pi}{4}$   
 (C)  $\frac{\pi}{6}$   
 (D)  $\frac{\pi}{9}$
42. Volume of the region  $S = \{(x, y, z) ; |x| + 2|y| + 3|z| \leq 6\}$  is  
 (A) 36  
 (B) 48  
 (C) 72  
 (D) 102

### Section – A (Maximum Marks: 24)

This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

43. The sum of smallest positive values of arguments of all the roots of the equation  $z^n = \lambda$ ,  $n \in \mathbb{N}$ ,  $\lambda \in \mathbb{R} - \{0\}$ , is even multiple of  $\pi$ , if  
 (A)  $\lambda < 0$ ,  $n$  is even  
 (B)  $\lambda < 0$ ,  $n$  is odd  
 (C)  $\lambda > 0$ ,  $n$  is even  
 (D)  $\lambda > 0$ ,  $n$  is odd

44. Deepak uses one of his four colored pens namely. Red, Green, Blue, Black for appearing in his examinations. He randomly uses one of the three pens which was not used in previous examination. If for his first exam he chooses the Red pen and  $P(K)$  denotes the probability that he chooses the Green pen in  $K^{\text{th}}$  examination, then
- (A)  $P(3) = \frac{2}{9}$
- (B)  $P(4) = \frac{7}{27}$
- (C)  $P(5) = \frac{7}{27}$
- (D)  $P(6) = \frac{61}{243}$
45. Let  $f(x)$  be a real valued function defined as  $f(x) = x^2 + 2x \int_{-2}^2 f(t) dt + 3x^2 \int_{-2}^2 f(t) t dt$ , then which of the following do/does not hold good?
- (A)  $\int_{-2}^2 t f(t) dt = -\frac{48}{503}$
- (B)  $\int_{-2}^2 t f(t) dt = -\frac{512}{503}$
- (C)  $f(2) + f(-2) = -\frac{1033}{503}$
- (D)  $f(2) - f(-2) = -\frac{192}{503}$
46. If  $f(x) = \text{sgn}((\cos^2 x + \cos x - 1)(\cos^2 x + \cos x + 1))$  has 6 points of discontinuity for  $x \in \left(0, \frac{n\pi}{2}\right)$ ,  $n \in \mathbb{N}$ , then
- (A) sum of digits of the sum of possible values of  $n$  is 5
- (B) sum of digits of the sum of possible values of  $n$  is 6
- (C) number of prime factors of the product of possible values of  $n$  is 5
- (D) number of prime factors of the product of possible values of  $n$  is 3
47. Consider the circle  $x^2 + y^2 - 12x + 4y + 15 = 0$ . Let  $C$  be the centre of the circle. Also, tangents at  $P(3, -6)$  and  $Q(10, -5)$  meet at  $R$ . Let  $S = 0$  represents family of circles passing through  $P$  and  $Q$ , then
- (A) area of quadrilateral  $CPRQ$  is 25 sq. units
- (B) the smallest possible circle of the family  $S = 0$  is  $x^2 + y^2 - 13x + 11y + 60 = 0$
- (C) the radical axis of the family of circles  $S = 0$  is  $x - 7y - 45 = 0$
- (D) sum of squares of abscissa and ordinate of point  $R$  is 145
48. If  $\cos^{-1}\left(\frac{x^2}{3}\right) + \cos^{-1}\left(\sqrt{1 - \frac{x^4}{9}}\right) + \cot^{-1} y = \frac{5\pi}{6}$ , then
- (A) maximum value of  $[|xy|]$  is 1
- (B) minimum value of  $[|xy|]$  is 1
- (C) maximum value of  $x^2 + y^2$  is  $\frac{1}{3}$
- (D) maximum value of  $x^2 + y^2$  is  $\frac{10}{3}$

**Section – B (Maximum Marks: 12)**

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

49. Let  $S = \sum_{k=1}^{624} \frac{1}{(\sqrt{k} + \sqrt{k+1})(\sqrt[4]{k} + \sqrt[4]{k+1})}$ , then S equals

50. If  $(1 + \lambda x + \mu x^2)^4 = k_0 + k_1 x + k_2 x^2 + \dots + k_8 x^8$ , where  $\lambda, \mu, k_0, k_1, k_2, \dots, k_8 \in \mathbb{R}$ .

Also, given  $k_0 + k_1 + k_2 \neq 0$  and  $\begin{vmatrix} k_2 & k_0 & k_1 \\ k_0 & k_1 & k_2 \\ k_1 & k_2 & k_0 \end{vmatrix} = 0$ , then the value of  $4\lambda + 32\mu$  is

51. Circle S is centred at (4, 0) with radius 2. Circle S' touches the circle S and also the y-axis. If the locus of the centre of the variable circle S' can be written as  $x = \frac{y^2}{\lambda} + \mu$ . Then find  $\lambda - \mu$

**Section – C (Maximum Marks: 12)**

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

**Question Stem for Question Nos. 52 and 53**

**Question Stem**

If  $\sum_{r=0}^{p_2} (p_1 r - 3) \cdot {}^{p_2}C_r = (2^{10})(38)$  where  $p_1, p_2 \in \mathbb{N}$ , then

52.  $p_1$  is equal to

53.  $p_2$  is equal to

**Question Stem for Question Nos. 54 and 55**

**Question Stem**

Suppose  $f(x)$  is a function satisfying the following conditions:

(i)  $f(0) = \frac{1}{2}$ ,  $f(3) = -\frac{13}{4}$ ; (ii)  $f(x)$  has a maximum value at  $x = -1$ ;

(iii)  $f'(x) = \begin{vmatrix} 2px & 2px-1 & 2px+q+1 \\ q & q+1 & -1 \\ 2(px+q) & 2px+q+1 & 2px+q \end{vmatrix}$ . For  $p_1 = f(-1)$  and range of  $f(x)$  as  $(-\infty, p_2]$ , then

54.  $p_1$  is equal to

55.  $p_2$  is equal to

**Question Stem for Question Nos. 56 and 57****Question Stem**

Let  $f(x) = \frac{x^4 - x^2}{x^6 + 2x^3 - 1}$ ;  $x > 1$ . If  $p_1 = p + q + 1$  and  $p_2 = abc$  where maximum values of  $f(x)$  is  $\frac{p}{q}$ , where  $p$  and  $q$  are natural numbers in their lowest form and extreme of  $f(x)$  occurs at  $x = \frac{a + \sqrt{b}}{c}$  where  $a$ ,  $b$  and  $c$  are relatively prime numbers pair-wise, then

56.  $p_1$  is equal to

57.  $p_2$  is equal to